

Krishna Harish
Lawrence E. Elkins High School, Missouri City, Texas, USA
krishnahaarish2009@gmail.com

Manuscript ID: ao-2026-07482d (revised submission)

Dear Prof. Skorb,

Please find enclosed the revised version of “Objective-Dependent Active Learning and Calibrated Uncertainty for Sample-Efficient Discovery of Rare-Earth Extractants: A Benchmark on 1,202 Experimental Distribution Coefficients with Prospective Validation” (ao-2026-07482d), together with a point-by-point response to the reviewer, a change-highlighted copy, and a clean copy.

We are grateful to the reviewer for a detailed and constructive report. We have addressed every point with new experiments on the real data; no claim is made without a computed result. The revision is substantial: the manuscript now reports five findings (adding a leakage-controlled generalization analysis and a definition-dependent discovery analysis), four new tables, and five new figures, all regenerated by released scripts. The principal additions, each responding to a specific reviewer request, are:

1. **Leakage-controlled evaluation** (Reviewer 1.1): recovering the ligand, source, and lanthanide grouping structure from the released features, we show that random row-level cross-validation ($R^2 = 0.88$) is strongly optimistic: ligand-, source-, and structural-family-disjoint splits collapse the estimate to at or below the mean. The in-distribution R^2 is now explicitly reframed as not a deployability estimate.
2. **Ligand-batch acquisition** (Reviewer 1.2): the realistic laboratory decision unit (select a ligand, measure all its conditions); the objective-dependent dichotomy persists.
3. **Definition-resolved discovery** (Reviewer 1.3): recall reported for rows, ligand–metal pairs, and unique ligands, with enrichment factor, regret, and hit-rate; the strategy ranking inverts across definitions.
4. **Rigorous subgroup calibration** (Reviewer 1.4): ligand-disjoint train/calibration/test; coverage by logD region and lanthanide class; Mondrian group-conditional conformal; and a demonstration that non-monotone calibration reorders the uncertainty ranking.
5. **Broader model and uncertainty-estimator baselines** (Reviewer 1.5): random forest, quantile regression forest, Gaussian process, gradient boosting, and an MC-dropout Bayesian neural network, plus a selectivity (separation-factor) analysis showing a high distribution coefficient alone is not sufficient.

We have also added figures and tables throughout (Reviewer 1.6, 1.7) and expanded the literature with 17 verified, up-to-date references (Reviewer 1.8).

All code, recovered group labels, and split definitions are released at <https://github.com/krishnatheaverage/logd-active-learning-benchmark> and archived on Zenodo; every reported number is reproducible on a single laptop. The manuscript is original and not under consideration elsewhere; use of an AI tool is disclosed within; the author has no competing interests to declare, and there are no funding sources to report.

Thank you for the opportunity to revise.

Sincerely,
Krishna Harish